Custom Scatter Plots

Custom scatter plots expose deeper insight into subgroups within a population, motivating students to define their own functions and deepen their analysis.

Lesson Goals	Students will be able to… • make custom scatter plots using helper functions
Student-facing Lesson Goals	• Let's explore a new Pyret function that lets us make more interesting scatter plots.
Prerequisites	• Simple Data Types • Contracts • Bar and Pie Charts • Probability, Inference, and Sample Size • Scatter Plots • Functions: Contracts, Examples & Definitions
Materials	• Lesson Slides • Custom Scatter Plot Starter File • Defining Row-Consuming Functions (Page 32) • Custom Scatter Plot - Explore (Page 33) • *Optional: Custom Scatter Plot with Piecewise Functions Starter File* • *Optional: Custom Animals Starter File* • *Optional: Dots for Value Ranges Starter File* • *Optional: species-dot - Explore*

Glossary

conditional :: a code expression made of questions and answers

data type :: a way of classifying values, such as: Number, String, Image, Boolean, or any user-defined data structure

piecewise function :: a function that computes different expressions based on its input

scatter plot :: a display of the relationship between two quantitative variables, graphing each explanatory value on the x axis and the accompanying response on the y axis

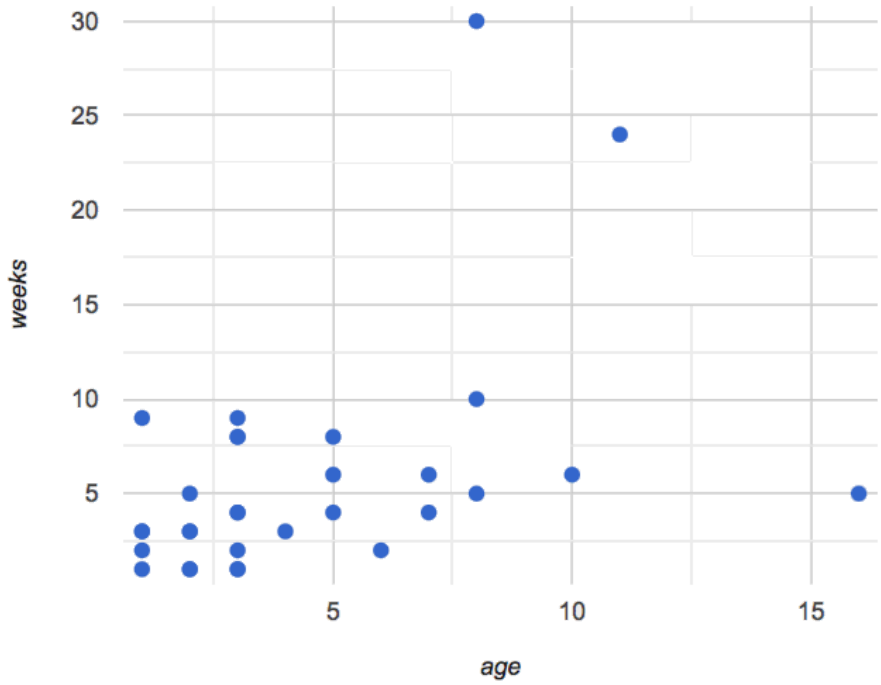
Image Scatter Plots *40 minutes*

Overview

Students are introduced to their first example of a *custom scatter plot* , which makes use of the `image-scatter-plot` function to replace each animals' dot with the name of the species. They then write code for another custom scatter plot.

Launch

Not all data points are created equal! Dogs and rabbits are very different, but when viewing all the animals on the same scatter plot, these differences are hidden.
This makes the displays more interesting, and helps us to dig deeper into analyzing the data.



Investigate

Let's start with some review, combining our knowledge of row and column lookups with our knowledge of how to define functions. *Note: Confirm that students' responses are correct before moving from one page to the next (below). Checking for understanding is crucial here.*

- Turn to Defining Row-Consuming Functions, and complete numbers 1-3 at the top of the page.
- When you are confident with the code you have written - check with your teacher or a partner - complete the rest of the page to define a function `species-tag` .
- Now, open Custom Scatter Plot Starter File. Does your definition of `species-tag` match what you see here?
- What do you Notice? What do you Wonder?
- What functions do you see defined? What do they do? What Rows do you see defined?

The contract for `image-scatter-plot` looks pretty different from other contracts we've seen.

```
image-scatter-plot :: (
  t  :: Table,
  xs :: String,
  ys :: String,
  f  :: (Row -> Image)
) -> Image
```

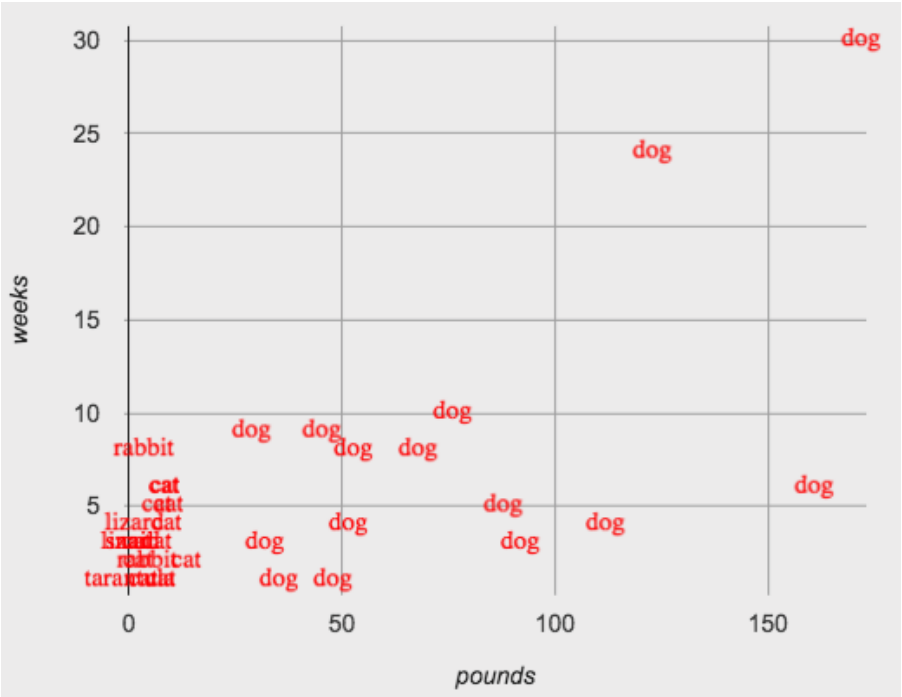
The function `image-scatter-plot` has an interesting Domain: *Table, String, String and… what is that last part?… **a Function that consumes a Row and produces an Image**!*

- Complete Custom Scatter Plot - Explore.
- As you work, consider how `image-scatter-plot` is different from other functions we've seen, and how it might be useful to our data analysis.

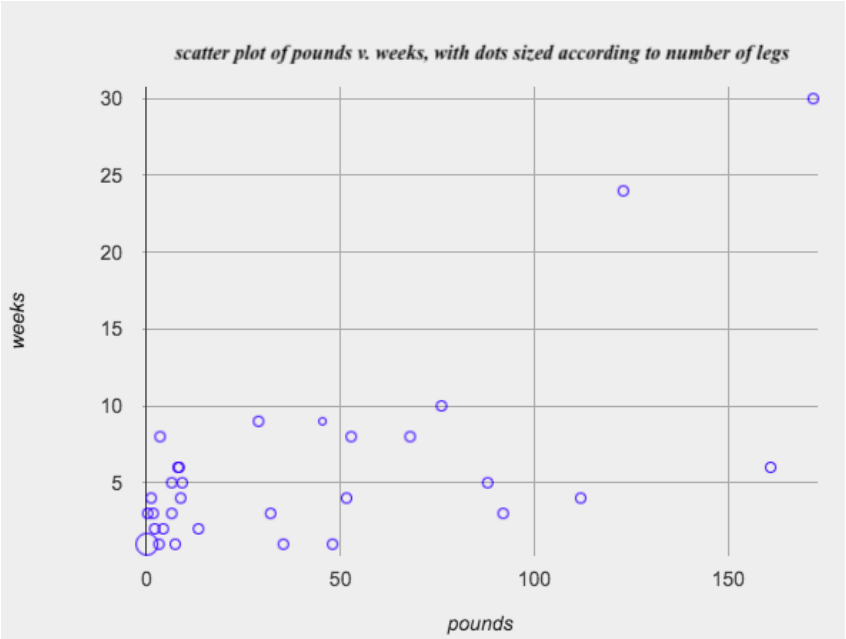
Synthesize

Ask students how having the dots labeled with the species changes their understanding of the data. Students should recognize that the `species-tag` ***scatter plot*** makes it clear that we may want to analyze each species separately, rather than grouping them all together. (In the Grouped Samples lesson, students will learn how to do just that!)

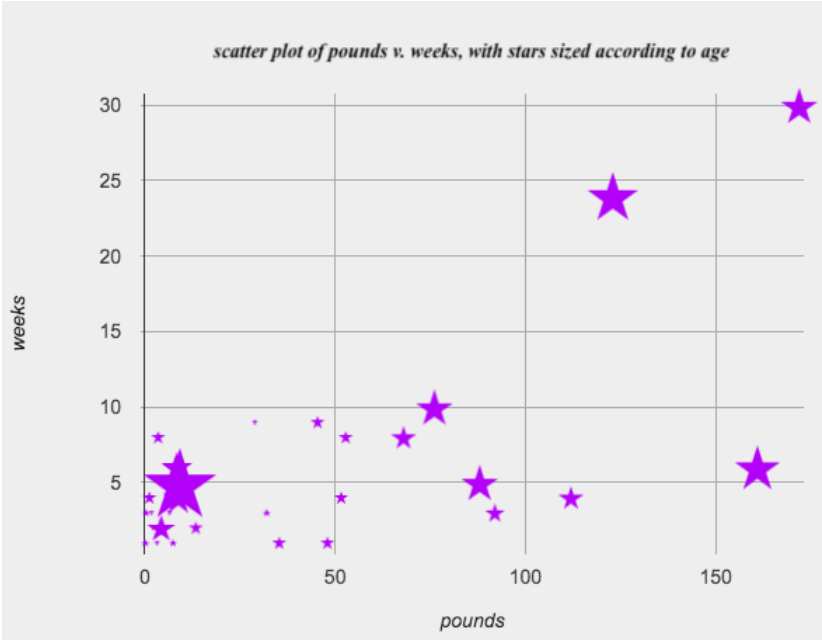
Ensure that students understand what the various functions in the file do. The `species-tag` function labeled dots with the species:



The `legs-tag` function made blue rings whose size was determined by the number of legs the animal had:



And the `age-tag` function made stars whose size was double the age of the animal:



- What *data type* is `species-tag`?
  - *A function that consumes a Row and produces an Image*
- How does `image-scatter-plot` work?
  - *It consumes a Table, a String, a String, and a Function that consumes a Row and produces an Image. It can label each point with some text, a different sized dot, or a star.*
- What are some other ways you could use columns of the Animals Dataset to display more interesting image-scatter-plots?
  - *Answers will vary.*
- What does an `image-scatter-plot` expression need to have defined in order to display a scatter plot with the animals' names in place of dots?

    - *We would need to define a* `name-tag` *helper function.*

To close this segment of the lesson, invite students to consider how `image-scatter-plot` might be useful to their *own* analysis.

Optional: Using Conditionals *25 minutes*

Overview

Students discover how to use ***conditionals*** - ***piecewise functions*** in math - to change dot colors and sizes, how "dot appearance" can be used to show more data in a scatter plot, and why that would be valuable.

NOTE: Math teachers may want their students to confront piecewise functions more deeply, and CS teachers may want to spend more time on conditionals. While not a part of the Data Science pathway, the Piecewise Functions and Conditionals lesson includes a lot of supporting material and practice pages for these topics, including the student notes on piecewise functions.

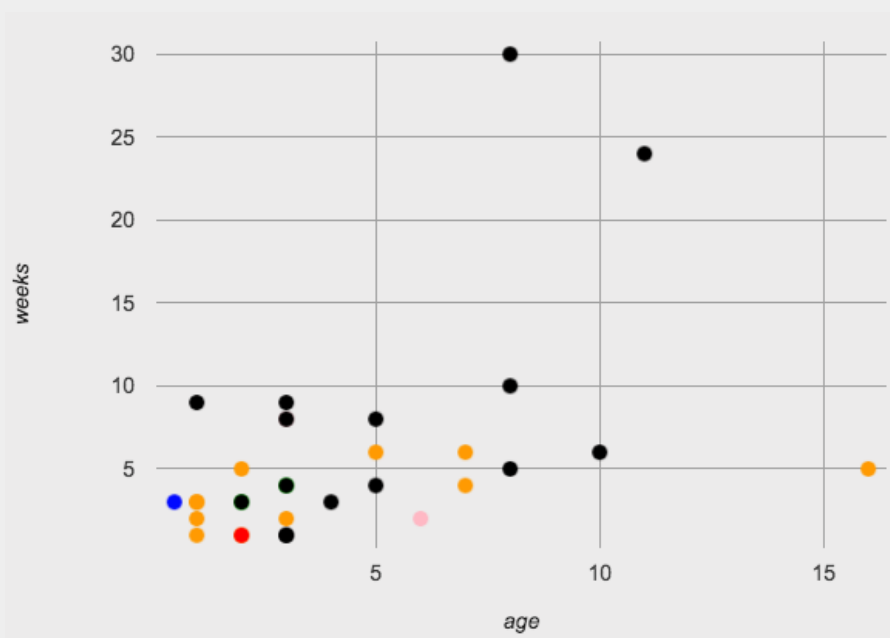
Launch

So far, we've seen that…

- the `scatter-plot` function makes uniform blue dots
- the `image-scatter-plot` function can label each point with some text, a different sized dot, or a star.

Explain to students that to get more out of the `image-scatter-plot` function, we'll need to use a different kind of function called a "piecewise function".

- Open the Custom Scatter Plot with Piecewise Functions Starter File and complete species-dot - Explore.



- What does this new visualization (above) tell us about the relationship between age and weeks?
  - *In general, as an animal gets older, the weeks to adoption increase. This appears to be true for both dogs (black dots) and cats (orange dots).*
- What other analysis would be helpful here?
  - *Sample answer: It might be interesting to look at the outliers to understand why some animals' adoptions take a longer amount of time.*

Investigate

- Open Word Problem: species-dot

- What is the contract for `species-dot`?
  - *The function name is* `species-dot`*, its Domain is Row, and its Range is Image.*
- What is the purpose of `species-dot`?
  - *It takes in a Row from the animals table and returns a solid, 5px circle using black for dog, orange for cat, pink for rabbit, red for tarantula, green for lizard and blue for snail.*
- How many examples do we need to write?
  - *We need to write six examples, one for each species.*
- Optional: Complete the Word Problem: sex-dot, to write a new helper function that will make differently-colored dots based on the animals' sex.

Make sure that students write the Contract *first*, and check in with their partner *and* the teacher before proceeding. Once they've got the Contract, have them come up with **`examples:`** for *each sex*. Once again, have them check with a partner *and* the teacher before finishing the page.
Debrief, and ask students to explain what the code does. Pay special attention to students' ability to articulate the "if/then" statements!

- Turn to Animal-Image - Explore and open the Custom Animals Starter File Starter File.
- How does using clipart help us to better understand the data?
  - *The images are cool! And they make it so easy to understand whether the species are evenly distributed or clustered.*
- What *risks* might there be to using clipart in displays?
  - *Sample response: Clipart of humans runs a serious risk of stereotyping or excluding populations!*
- We have seen a lot of different image scatter plots today! What ideas do you have for how `image-scatter-plot` could be used to deepen the analysis of your dataset?

Optional: When your conditional is *already* a Boolean

If you have time or students who are ready for a challenge, you can also have them make a scatter plot for dots distinguishing whether the animal is fixed or not using the directions at the end of the starter file or Word Problem: fixed-dot. Students will discover that this is a little different from the other two functions they've seen because `fixed` is already a Boolean column! The code will work if written in either of the following ways:

Checking the Boolean	Using the Boolean Directly

`fun fixed-dot(r):`
`  if  (r["fixed"] == true) : circle(5, "solid", "green")`
`  else if (r["fixed"] == false): circle(5, "solid", "black")`
`  end`
`end`
	`fun fixed-dot(r):`
	`  if r["fixed"]: circle(5, "solid", "green")`
	`  else: circle(5, "solid", "black")`
	`  end`
	`end`

For students who are really ready for a challenge, direct them to the Dots for Value Ranges Starter File and Dots for Value Ranges - Explore

Synthesize

How do piecewise functions expand what is possible with the `image-scatter-plot` function?